



DPUSGETextureNDI

Creates a dynamic video texture in the virtual world. The texture contains an image that is received over the network from a NDI server, such as DPUSGEScreenServerNDI. Possible data format is RGBX, RGBA, BGRX, BGRA and UYVY.

1. Static parameters:

<i>View</i>	Name of the SGE view for which the dynamic texture is updated. Additionally, it will be visible in all views that share textures with the specified SGE view. Default: <i>SGEView</i>
<i>TextureName</i>	Name of the created video texture, which must start with a dollar sign (\$). Default: <i>\$Video</i>
<i>Source</i>	Source name of the stream. In NDI, the name of the source is <Computer> (<NameofSource>)
<i>ConnectTimeoutMS</i>	Timeout in milliseconds for the connection to the ScreenSend server. Default: 3000
<i>W, H</i>	Size of the size of each frame of the Stream. If W and H is used, the DPU starts immediately and does not wait for a server connection. Otherwise, the DPU will wait for the connection to the server to get the size of the first frame. Once the size has been determined, only frames that fit the size are accepted.

2. Outputs:

<i>ImageW, ImageH</i>	Image size (in pixels) of the received image.
<i>FPS</i>	Received image frame rate.
<i>FrameNumber</i>	Frame number of the last received image.

3. Functional principle:

Using *DPUSGETextureNDI*, a dynamic video texture is created in the virtual world. The texture contains a live image that is received from a NDI source over the network.

For example, when combined with *DPUSGEScreenServerNDI*, the graphical view of one computer can be embedded into another's.



4. Example of a SGE object which references a video texture:

```
define SIG3DObject VideoObj
{
    AlphaTest = 0.2;
    Blend = false;
    // [...]
    Texture = $MyVideo;
    TextureFilter = NoCompression | ClampS | ClampT | MagLinear
                  | MinLinear | Tex-ture;
    // [...]
};
```

5. Dependencies to other DPUs:

Please make sure that the computer on which *DPUSGETextureNDI* runs also executes *DPUSGEWorld*, *DPUSGEObjectXXX* (stands for any DPU that uses the dynamic video texture in one of its displayed objects) and *DPUSGEView*. The *Index* values of these DPUs must be set in the following order:

DPUSGEWorld.Index < *DPUSGETextureNDI.Index* < *DPUSGEObjectXXX.Index* < *DPUSGEView.Index*

6. Example:

This DPU works well using the following script:

```
DPUSGETextureNDI VidTexture
{
    Computer = {VIS};
    Index = 50;
    Source = "VISFRONT (SGEView)";
    # VISFRONT is the hostname of the computer
    TextureName = "$MyVideo";
};
```